

Hotel Booking System Project Plan

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Spring SDEV 265

Professor Parrot

February 15, 2026

Project Plan (Hotel Booking System)

Introduction(Frances)

This project is a hotel reservation management system that will allow users to store information about reservations, check in and check out guests, and search for reservations. Hotel reservation systems are typically complex, so we aim to produce a more user-friendly version. One of our team members is a current employee of a hotel, with over a year of experience working with various reservation systems. We will leverage this expertise to curate a system that meets the unique needs of both hotel employees and guests. Another important aspect of hotel software is privacy and security. Hotels require important information, such as credit card numbers and addresses, and it is crucial to ensure that guests are confident that their information is stored securely and responsibly.

Project Organization(Gentry)

<u>Position</u>	<u>Names</u>	<u>Description</u>
Organizer	Gentry Bayer	Ensures that the team meets the required task to complete the project and keeps all materials in check.
Designer	Frances Jones	Ensures that the software has a simple design that is easy for the project team to implement.
Programmer	Gentry Bayer	Writes code for the program.
Programmer	Frances Jones	Writes code for the program.
Tester	Gentry Bayer	Analysis ensures that the program does not have any flaws in its design and runs smoothly.

Risk Analysis(Frances)

Data Security/Privacy Issues	The program will store sensitive information like names, addresses, and payment information. Hotels are common targets for identity theft, so security is important.	High Risk	- Use hashed passwords for all users. - Practice secure database techniques. - Use the principle of least privilege. - Validate all user inputs.
Simulated Payment Issues	Since we can't actually collect payments, we will have to simulate them. This creates a risk of validation issues.	Medium Risk	- Validate card number length and format. - Validate expiration date and format. - Test multiple scenarios.
Schedule Delays	Various stages may take longer than expected due to unexpected issues like bugs or integration issues.	Medium Risk	- Follow the project schedule. - Break larger tasks into small chunks. - Communicate often between team members.
Software Integration Issues	There may be issues connecting the GUI, database, and backend.	Medium Risk	- Test everything individually first. - Use version control (GitHub). - Maintain consistent coding standards.
Hardware/Software Compatibility Issues	The program may behave differently across operating	Low Risk	- Test on multiple OS. - Standardize the

	systems and/or Python versions		development environment. - Make setup clear.
UI Usability Issues	Users may struggle with the program if it is confusing or difficult to use.	Low Risk	- Make interface simple and consistent. - Perform testing.
Data Loss/Database Issues	Database issues or accidental deletion could mean the hotel loses reservations or other data.	Low Risk	- Enforce regular backups of the database. - Test database recovery procedures.

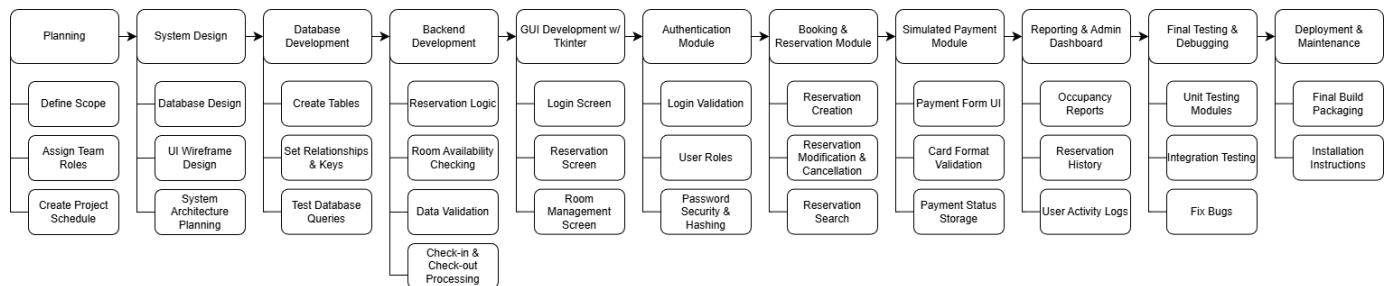
Risk**Description****Risk Level****Mitigation****Hardware/Software Requirements(Gentry)****Hardware Requirements**

- A Windows or Mac computer
- Monitor or laptop screen
- Keyboard and mouse

Software Requirements

- Operating System: Windows 10/11 or macOS (latest versions recommended)
- Python (version 3.8 or higher)
- Tkinter (Python GUI library, comes pre-installed with Python)
- Visual Studio Code (or any preferred Python IDE)
- SQL Database (e.g., SQLite, MySQL, or PostgreSQL)

Work Breakdown (UML Diagram)(Frances)



Project Schedule(Gentry)

Phase 1: This is the planning and design phase of the project.

- (T1)Planning the project- Week 1
- (T2)Designing the system- Week 2
- The design of the system will depend on how the project planning goes.

Phase 2: This is the main development part of the project.

- (T3)Setting up the database using SQL- Week 3-4
- (T4)Starting the backend development in Python- Week 3-5
- (T5)Doing the GUI development using Tkinter in Python- Weeks 3-5
- These parts in this phase of the project might occur simultaneously due to them all depend on the design of the system.

Phase 3: This is the implementation of the modules part of the project.

- (T6)Import the user authentication module- Weeks 6-8
- (T7)Import the room management module- Weeks 6-8
- (T8)Import the booking and reservation module- Weeks 6-8
- None of this can occur unless the backend and gui development is complete.

Phase 4: This is the administrative and payment features phase of the project.

- (T9)Payment processing- Weeks 8-10

- (T10)Reporting and admin dashboard- Weeks 8-10
- The payment processing will depend on the booking module being completed.

Phase 5: This is the testing and deployment phase of the project.

- (T11)Testing and debugging- Weeks 10-14
- (T12)Deployment and maintenance of the project- Weeks 10-14
- The testing cannot be done until all modules are completed, and the deployment and maintenance cannot be done until the testing of the project is finished.

Monitoring and Reporting Activities(Frances)

We will be using GitHub to track changes and as version control. We will provide comprehensive commit messages to communicate any changes made, along with regular team meetings to discuss progress and any issues that arise. We will work to have thorough documentation of the entire project.

Post-Project Review:

During the course of the project we decided some things in our original plan were out of scope, like payment processing and administrative privileges. However, besides that, our workflow ended up matching our plan very closely.

Appendix (Activity Table)(Gentry)

<u>Task</u>	<u>Effort (Persons)</u>	<u>Duration (Weeks)</u>	<u>Dependencies</u>
(T1) Planning the project	1	1	N/A
(T2) System design (Database + UI Design)	1	1	T1
(T3) Database setup (SQL Schema & Tables)	2	2	T2
(T4) Backend development (Python logic)	2	3	T2
(T5) GUI development (TKinter Interface)	2	3	T2
(T6) User Authentication Module	1	2	T3-5
(T7) Room management Module	2	2	T3-5
(T8) Booking and Reservation Module	2	2	T3-5
(T9) Payment Processing for the system	1	2	T8
(T10) Reporting & Admin Dashboard	2	2	T6-8

(T11) Testing and Debugging	2	2	T3-10
(T12) Deployment and Maintenance	2	2	T11

End of Project Plan